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Integrated modelling and simulation of liquefied CO_2 tank and cargo handling system with new marine transportation greenhouse gas intensity

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ABSTRACT

Ship transport of liquefied carbon dioxide (LCO_2) is competitive for long-distance transportation in the supply chain of carbon capture and storage, but inevitably has a problem of pressurization due to boil-off gas (BOG) generation from the low-temperature LCO_2. Conventionally, BOG can be vented, but it results in increased greenhouse gas (GHG) emissions. This study suggests two advanced cargo handling systems (CHS) of continuously and periodically operated re-liquefaction for handling the LCO_2 BOG, and compares the GHG intensity with that of the BOG venting case. A new index of marine transportation GHG intensity (MGI) was developed to accurately evaluate the well-to-wake GHG intensity of shipping and advanced CHS. An integrated model of the tank and CHS was built to evaluate their continuous interaction. The results with BOR_0 (design specification of boil-off rate) of 0.15 % show that the venting case has a lower MGI up to 13 days of short voyage, but the re-liquefaction systems are preferable for longer voyages because vented CO_2 increases faster in the venting case. Additionally, the periodically operated re-liquefaction system can be considerable for a wide range of voyage days, as it enables flexible operation to cover shorter voyages with low BOR_0.

Nomenclature

Q	heat (W)
q	heat flux (W/m^2)
f	heat distribution factor
h	heat transfer coefficient (W/(m^2·K))
A	area (m^2)
T	temperature (°C)
P	pressure (bar)
PW	power consumed by compressor (kW)
$\dot{m}$	mass flow rate (kg/h)
$\dot{n}$	molar flow rate (kgmole/h)
n	molar flow (kgmole)
ρ	mass density (kg/m^3)
L_v	latent heat of vaporization (kJ/kg)
V	volume (m^3)
Nu	Nusselt number
Pr	Prandtl number
Ra	Rayleigh number
k_{th}	thermal conductivity (W/(m·K))
FR	filling ratio (%)
t	time (h)

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BOR	Boil-off rate (%/day)
MGI	Marine transportation GHG Intensity (g_{CO2eq}/(t·nm))
GFI	GHG fuel intensity (g_{CO2eq}/MJ)
EEDI	Energy Efficiency Design Index (g_{CO2}/(t·nm))
X	Reduction factor (%)
GWP	Global warming potential
CF_j	j emission conversion factor for the combustion of the fuel (g_j/g_{fuel})
SFC	Specific fuel consumption (g_{fuel}/kWh)
FC	Fuel consumption (g_{fuel}/h)
V_{ref}	Reference velocity (knots)
MCR	Maximum continuous rating (kW)
LCV_k	Lower calorific value (MJ/g_{fuel})
$GHG_{k,WtT}$	Well-to-tank GHG intensity of fuel k (g_{CO2eq}/MJ)
Subscripts	
v	vapor
l	liquid
if	interface
a	ambient
0	initial
vi	vapor-interface

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(*continued*)

il	interface-liquid
liq	liquefied
tot	total
lam	laminar
turb	turbulent
Voy	voyage
j	type of greenhouse gas
k	type of fuels
Superscripts	
i	type of systems
avg	average
Abbreviations	
CO_2	Carbon dioxide
CCUS	Carbon Capture, Utilization & Storage
GHG	Greenhouse gas
LCO_2	Liquefied CO_2
BOG	Boil-off gas
CHS	Cargo handling system
LNG	Liquified natural gas
N_2	Nitrogen
WtW	Well-to-Wake
WtT	Well-to-Tank
TtW	Tank-to-Wake
IMO	International Marine Organization
TMZM	Thermal Multi-Zone Model
PSO	Particle Swarm Optimization
ME	Main engine
CAE	Conventional auxiliary engine
VENT	Venting system
NCAE	Non-conventional auxiliary engine
RLIQ	Re-liquefaction system

1. Introduction

Carbon dioxide (CO_2) transportation has gained considerable research attention because of the increasing global demand for carbon capture, utilization, and storage (CCUS). At the G7 summit held in May 2023, an agreement was reached regarding the importance of the role of CCUS in reducing greenhouse gas (GHG) emissions from the industry. There has been an increase in the number of CCUS projects as a response to this global demand. The CO_2 transportation infrastructure enables the flexible development of CCUS projects and facilitates effective scale-up by connecting projects that focus on capture or storage [1]. Consequently, technologies that enable the large-scale transport of CO_2 have been actively developed.

The transport of liquefied CO_2 (LCO_2) stored under low pressures via ships is considered a competitive approach for long-distance transportation. Ship transportation is an economical option and can achieve flexible transportation because it does not require infrastructure for long pipelines. In countries with limited domestic storage capacity, CO_2 transportation over long distance is inevitable, emphasizing the need for developing large CO_2 carriers. LCO_2 is preferred for transportation by ship because LCO_2 at 7.5 bar and −47 °C is 630 times denser than gaseous CO_2 at room temperature and pressure. This difference in density caused by phase change makes it advantageous to transport CO_2 in the liquid state. LCO_2 can be transported at low (<10 bar) and medium storage pressures (>10 bar). Jeong et al. [2] showed that low storage pressure of 6.5 bar has an economic advantage over a medium storage pressure of 15 bar. HD Hyundai, which is building a large LCO_2 carrier, also revealed that it is building ships that store LCO_2 at a low pressure of 5.5 bar [3].

LCO_2 needs to be stored below atmospheric temperature, and therefore, boil-off gas (BOG) generation should be handled by a cargo handling system (CHS) similar to that for other low-temperature fluids such as liquified natural gas (LNG). The influx of heat into the cargo tank is the primary factor contributing to BOG production; this influx occurs because of the difference between the ambient temperature and the temperature of the stored cargo. The tank pressure increases gradually because the BOG is continuously generated in the cargo tank. LCO_2 can be transported without considering BOG handling during the voyage if the pressure increase is within the allowable operating pressure of the tank. However, the pressure may increase above the allowable pressure with an increase in the number of voyage days, resulting in safety issues and requiring a cargo handling strategy. The first option is venting the excess BOG which is necessary to maintain the tank pressure within the allowable pressure range; however, this causes not only a loss of cargo but also increases GHG emissions from LCO_2 carriers because the vented cargo is CO_2. The second option is installing a BOG re-liquefaction system that can manage the tank pressure without venting the BOG. The operation of the BOG re-liquefaction system consumes energy, and therefore, GHG emissions occur because of fuel consumption. Therefore, a cargo management strategy must be determined based on a quantitative comparison between the GHG emissions of venting and reliquefying BOG based on the shipping scenario.

Several studies have been conducted on LCO_2 BOG re-liquefaction systems. Lee et al. [4] designed a CO_2 BOG re-liquefaction process using boil-off CO_2 as refrigerant and performed energy optimization. Lu et al. [5] designed a CO_2 BOG re-liquefaction system using ammonia as refrigerant and performed economic optimization. Yoo [6] designed and optimized a CO_2 BOG re-liquefaction system using LNG as the refrigerant. However, these studies assumed pure CO_2 and neglected the effects of impurities on system performance. The presence of impurities, even in small amounts, changed the phase equilibrium conditions, which affected tank pressure. Further, impurities reduce the performance of the re-liquefaction system, and therefore, the impact of impurities must be considered when targeting tanks for storing captured CO_2 [7]. Some studies considered impurities, e.g., Jeong et al. [2]; however, they made simplified assumptions about the BOG composition such as fixed nitrogen (N_2) concentrations of 5 or 15 %. The interaction between the cargo tank and BOG re-liquefaction system needs to be studied for a more accurate prediction because the concentration of impurities in the tank varies with time owing to heat ingress and the operation of the CHS.

Combined modelling of the tank and CHS is required because the tank and CHS continuously affect each other. Thus far, there have been some studies on BOG re-liquefaction systems; however, most research focused only on reducing the specific energy consumption at the system boundary of the re-liquefaction cycle [2,4–6]. Jeong et al. [2] proposed a re-liquefaction process using mixed refrigerants; however, the study considered only the fixed conditions of the BOG inlet and outlet and did not consider the effect of reinjected BOG into the tank. The tank and re-liquefaction system constantly affect each other. The thermodynamic state of the tank also changes when the reliquefied CO_2 is returned to the tank, resulting in a change in the BOG inlet conditions such as temperature, pressure, and composition. Therefore, developing models to determine these chain effects is necessary.

In addition, a new comprehensive index is required to estimate the well-to-wake (WtW) GHG intensity from shipping. A GHG fuel intensity (GFI), approved at International Maritime Organization (IMO) MEPC 83, represents WtW GHG emissions per unit of energy used on board [8]. A problem with GFI, however, is that it can misrepresent actual GHG emissions, for instance, reducing energy consumed on board can increase GFI. Another existing index for ships, the energy efficiency design index (EEDI) by the IMO, is assessed as the amount of CO_2 emissions generated when transporting one ton of cargo 1 nautical mile [9]. However, the scope of the IMO-defined index is currently limited to tank-to-wake (TtW) emissions. Further, the calculation methodology is propulsion-focused, which is not suitable to assess emissions from non-conventional systems such as BOG venting and BOG re-liquefaction systems. Therefore, extending the scope of the index is necessary to assess the entire transportation process by considering the life cycle of the fuel.

This study aims to design an LCO_2 CHS considering the effect of N_2 impurities and determine the cargo handling strategy by voyage days in terms of a new GHG intensity index. To this end, this study establishes a new index for ships, which is referred to as the marine transportation

GHG intensity (MGI). This can assess the WtW GHG intensity generated during the entire transportation process. A cargo tank model is built to analyze the tank behavior of the BOG generation and pressure build-up. Further, BOG venting and re-liquefaction systems integrated with the tank model are designed, and then, the optimal operating conditions of the BOG re-liquefaction system with minimum MGI are determined. Finally, this study suggests the best cargo handling strategy based on MGI analysis by voyage days.

2. Background knowledge

2.1. Energy efficiency design index

As defined by the IMO, EEDI represents the amount of CO_2 emissions when transporting 1 ton of cargo 1 nautical mile [9]. Attained EEDI represents the EEDI value achieved by an individual ship, whereas the required EEDI is the maximum attained EEDI value regulated for each type and size of ship. According to the EEDI regulation, the delivery of a ship is prohibited if its attained EEDI exceeds the required EEDI. The required EEDI is calculated by applying a reduction factor to the reference line value as shown in Eq. (1).

$$\text{Required EEDI} = \left(1 - \frac{X}{100}\right) \cdot \text{Reference line value} \tag{1}$$

Where X represents the reduction factor. The reduction factor for gas carriers is 30 % from 2025 [9]. As criteria for setting regulatory values had to be established, the IMO investigated CO_2 emissions from ships with a gross tonnage of 400 or more, built between 1999 and 2009, categorized by ship type. Then, regression analysis was used to define a reference line for each ship type. The reference line value is calculated using Eq. (2), where parameters a and c represent values determined by the ship type and b corresponds to the deadweight of the ship. In the case of the gas carrier, a and c were 1120.00 and 0.456, respectively [9].

$$\text{Reference line value} = a \cdot b^{-c} \tag{2}$$

2.2. Thermal multi-zone model of cryogenic tanks

The thermal multizone model (TMZM) is a part of a nonequilibrium model that compensates for the limitation that the equilibrium model oversimplifies the heat and mass transfer between the vapor and liquid [10]. The TMZM segments the tank into several zones such as the vapor, vapor–liquid interface, bulk liquid, wall, and boundary layer [11–13]. The TMZM assumes that the interface is a thin film of negligible mass. The vapor, interface, and liquid states are superheated, saturated, and subcooled, respectively, based on the measured temperature distribution (from the bottom to the top) in the self-pressurization test [13,14]. The model assumes that the temperature difference between the vapor and liquid forms a buoyant flow and induces natural convection between the vapor, interface, and liquid.

In the TMZM, the heat entering the tank is calculated as follows. The formulations for Q_v and Q_l are expressed as Eq. (3).

$$Q_{tot} = Q_v + Q_l = q_v \cdot A_v + q_l \cdot A_l \tag{3}$$

where q_v and q_l represent the heat fluxes into the vapor and liquid, respectively, and A_v and A_l represent the lateral areas of the vapor and liquid, respectively.

To describe the proportion of heat leaking into the vapor and liquid, the definition of heat distribution factor (f) is given by Eq. (4).

$$f = \frac{q_v}{q_l} \tag{4}$$

The description of Q_{vi} and Q_{il} are respectively described as Eqs. (5) and (6).

$$Q_{vi} = h_{vi} A_{vi} \left(T_v - T_{if}\right) \tag{5}$$

$$Q_{il} = h_{il} A_{il} \left(T_{if} - T_l\right) \tag{6}$$

where h_{vi} and h_{il} represent the heat transfer coefficients, and A_{vi} and A_{li} represent interfacial areas at the liquid level of the tank, respectively.

2.3. Conventional BOG re-liquefaction process

For CO_2 BOG re-liquefaction, previous studies have shown that using an external refrigerant offers energy advantages. Table 1 lists the literature on CO_2 BOG re-liquefaction system. Lee et al. [15] conducted an energy and economic evaluation of the Linde–Hampson BOG re-liquefaction system, vapor-compression refrigeration BOG re-liquefaction system under storage pressure of 7.01 bar. According to Lee et al., the vapor-compression re-liquefaction system using a pre-cooler achieves over 15 % reduction in specific energy consumption (SEC) compared to other systems while maintaining a similar level of total annualized cost. Furthermore, Chu et al. [16] and Jeong et al. [17] showed that the precooler using the external refrigerant can reduce SEC compared to Yoo's case [6] without the precooler, despite the presence of impurities.

Fig. 1 shows the conventional BOG re-liquefaction process using a refrigerant. Inlet BOG is compressed via the BOG compressor, then cooled to ambient temperature using air or water. The BOG undergoes heat exchange with an external refrigerant of a lower temperature before expanding through a Joule–Thompson (J–T) valve. By selecting proper J–T expansion path, the temperature of the BOG can be dropped and liquefied. The refrigerant is compressed, cooled, and expanded to make low temperature via a vapor compression refrigeration cycle, as shown in Fig. 1.

2.4. Particle swarm optimization

Particle swarm optimization (PSO) is a swarm-based stochastic optimization technique proposed by Eberhart and Kennedy [18,19]. PSO is part of an evolutionary algorithm and has strong distributed capabilities, which makes it suitable for nonlinear optimization and parallel computation.

Fig. 2 shows the PSO algorithm [20]. When optimization begins, a population of individuals, called particles, moves step by step across a region. Each particle updates its optimal position, and the optimal position of the population is the sum of all optimal positions experienced by the individuals. After finding the best position, it determines a new velocity for each particle. As they move, the evaluation is performed again, with each particle heading towards the best position it has found so far, while also heading towards the best positions found by other particles in the swarm. The optimal solution is obtained when the termination condition is satisfied after the iterations.

Table 1
Literature on the CO_2 BOG re-liquefaction system.

Feed concentration (mol%)	Storage pressure (bar)	SEC (kWh/ton$_{CO2}$)	Refrigerant	Reference
CO_2 100 %	7.01	99.3 84.2	– Ammonia	[15]
CO_2 100 %	6.59	92.7	–	[6]
CO_2 98.4 %	6.2	91.7	Ethane, Propane	[16]
CO_2 95 %, N_2 5 % CO_2 85 %, N_2 15 %	6.5	82.9–87.5 109.3–380.0	Ammonia	[17]

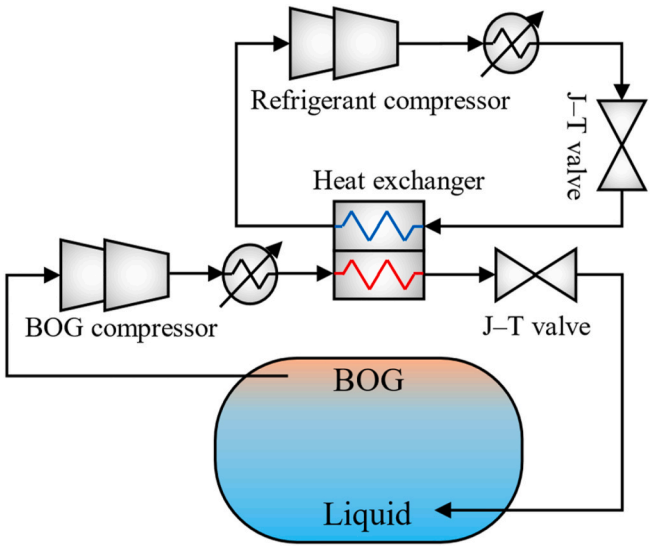


Fig. 1. Process flowchart of conventional BOG re-liquefaction process using refrigerant.

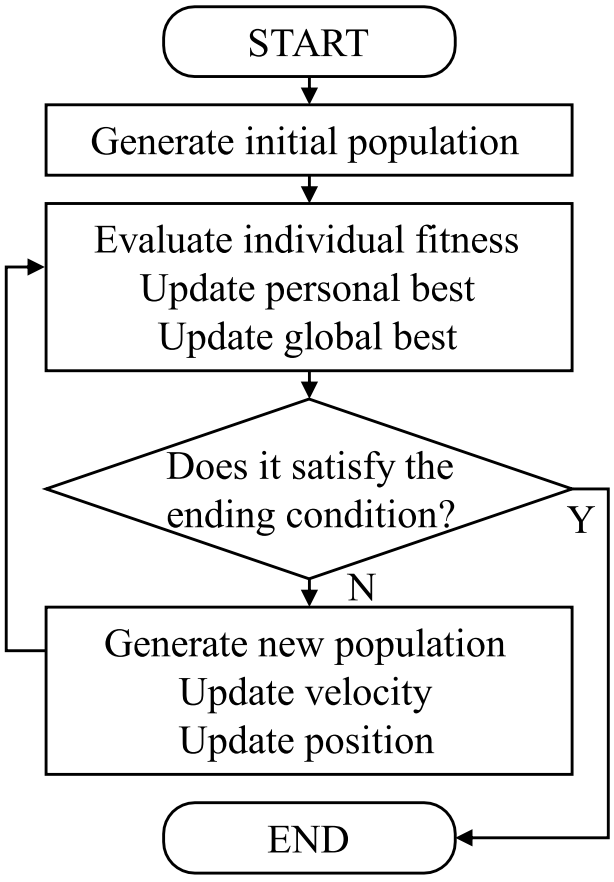


Fig. 2. Flowchart of the PSO algorithm.

3. Methodology

3.1. Marine transportation GHG intensity (MGI)

This study suggests a new index, MGI, which is defined as an extension of the existing EEDI to assess the WtW GHG intensity of maritime transportation as shown in Fig. 3. The MGI consists of WtW GHG emissions, including WtT and TtW GHG emissions. As shown in Fig. 3, the WtT GHG emissions include all GHG emissions generated during the production process of the used fuel, such as extraction, processing, and transportation. The TtW process in MGI includes GHG emissions from not only the propulsion and conventional auxiliary systems (CAE), but also non-conventional auxiliary systems (NCAE) such as re-liquefaction systems in CHS. In this study, three different cases of CHS were simulated to maintain tank pressure within allowable limits during a voyage: BOG venting system (Case 1) and BOG re-liquefaction system in continuous operation (Case 2) and periodic operation (Case 3). Thus, MGI's TtW GHG emissions include not only those from fuel combustion, but also those released directly into the atmosphere, such as those generated by the BOG venting system.

The MGI is defined as the GHG emissions from WtT and TtW processes per transport work (multiplication of unit cargo mass (tonne) and transport distance (nm)) as shown in Eq. (7). GHG emissions are calculated according to Eqs. (8) and (9) for WtT and TtW, respectively, using the amount of fuel consumed by each system. For WtT GHG emissions assessment, the WtT GHG emission per unit energy for fuel k ($GHG_{WtT,\,k}$) provided by FuelEU database was used, as shown in Eq. (8) [21]. To evaluate TtW GHG emissions, GHGs such as methane (CH_4) and nitrous oxide (N_2O) generated by fuel combustion are converted into CO_2 equivalent using their global warming potential (GWP), as shown in Eq. (9). Fuel consumed by each system is calculated using the power consumed as in Eq. (10).

$$\mathrm{MGI} = \frac{\sum_i \dot{m}^i_{\mathrm{WtT}} + \sum_i \dot{m}^i_{\mathrm{TtW}}}{\mathrm{Capacity}\cdot V_{\mathrm{ref}}} \tag{7}$$

$$\dot{m}^i_{\mathrm{WtT}} = \sum_k \left(\dot{\mathrm{FC}}^i_k \cdot \mathrm{LCV}_k \cdot \mathrm{GHG}_{\mathrm{WtT},k}\right) \tag{8}$$

$$\dot{m}^i_{\mathrm{TtW}} = \begin{cases} \sum_k \left\{ \dot{\mathrm{FC}}^i_k \cdot \sum_j \left(\mathrm{CF}_{k,j}\cdot \mathrm{GWP}_j\right) \right\} \\ \sum_j \left(\dot{m}^i_j \cdot \mathrm{GWP}_j\right) \text{ if } i = \text{venting systems} \end{cases} \tag{9}$$

$$\dot{\mathrm{FC}}^i_k = \begin{cases} \mathrm{PW}^i_k \cdot \mathrm{SFC}^i_k \\ 0 \text{ if } i = \text{venting systems} \end{cases} \tag{10}$$

where, $\dot{m}$ is mass flow of GHG emissions, $\dot{\mathrm{FC}}$ is mass flow of fuel consumptions, PW is power consumed by system, i is type of systems, j is type of GHGs and k is type of fuels.

Power consumed by main engine (ME) ($\mathrm{PW}^{\mathrm{ME}}$) and CAE ($\mathrm{PW}^{\mathrm{CAE}}$) are calculated according to the method of original EEDI, as shown in Eqs. (11) and (12), respectively [22]. The GHG emissions of the NCAE are calculated using the average value of the simulated values over the voyage days. For re-liquefaction systems that emit GHGs due to power consumption, the average power consumption ($\mathrm{PW}^{\mathrm{RLIQ,avg}}$) is calculated as in Eq. (13). For venting systems, the average amount of vented GHGs to the atmosphere is calculated as in Eq. (14).

$$\mathrm{PW}^{\mathrm{ME}} = 0.75 \cdot \mathrm{MCR}^{\mathrm{ME}} \tag{11}$$

$$\mathrm{PW}^{\mathrm{CAE}} = \begin{cases} 0.025 \cdot \mathrm{MCR}^{\mathrm{ME}} + 250 \text{ where, } \mathrm{MCR}^{\mathrm{ME}} \geq 10,000 \text{ kW} \\ 0.05 \cdot \mathrm{MCR}^{\mathrm{ME}} \text{ where, } \mathrm{MCR}^{\mathrm{ME}} < 10,000 \text{ kW} \end{cases} \tag{12}$$

$$\mathrm{PW}^{\mathrm{RLIQ,avg}} = \frac{\int_0^{t_{\mathrm{Voy}}} \mathrm{PW}^{\mathrm{RLIQ}} dt}{t_{\mathrm{Voy}}} \tag{13}$$

$$\dot{m}^{\mathrm{VENT,avg}} = \frac{\int_0^{t_{\mathrm{Voy}}} \sum_j \dot{m}^{\mathrm{VENT}}_j \cdot \mathrm{GWP}_j dt}{t_{\mathrm{Voy}}} \tag{14}$$

where t_{Voy} is ship's voyage duration.

Table 2 lists the parameters used in the MGI calculation in this study.

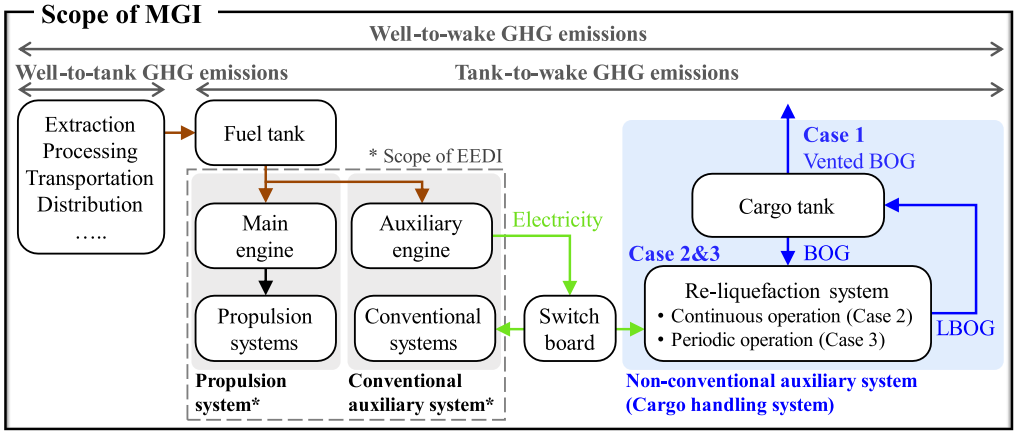


Fig. 3. Scope of MGI in this study.

Table 2
Parameters used to calculate MGI.

Item		Unit	Value	Reference
Reference value	SFC	g/kWh	136	[22]
	LCV_{LNG}	MJ/g	0.05	[21]
	$GHG_{WtT,LNG}$	g_{CO2eq}/MJ	18.5	
	$CF_{CO_2,LNG}$	g_{CO2}/g_{fuel}	2.75	
	$CF_{N_2O,LNG}$	g_{N2O}/g_{fuel}	0.00011	
	GWP_{CO_2}	–	1	
	GWP_{N_2O}	–	298	
Calculated value	MCR^{ME}	kW	21,455	Calculated based on IMO
	V_{ref}	knots	17.99	[23]

The MGI was calculated for ships using LNG as fuel for this study. Therefore, as listed in Table 2, the default emission factors for the LNG pathway were used as parameter values [21]. The maximum continuous rate of ME (MCR^{ME}) and reference velocity (V_{ref}) are calculated according to the rules of IMO [23]. Capacity is the final amount of CO_2 transported after the voyage days.

The MGI of a specific CHS, such as a BOG venting system or a BOG re-liquefaction system can be obtained using Eqs. (15) and (16).

$$MGI^{VENT} = \frac{\dot{m}_{WtT}^{VENT} + \dot{m}_{TtW}^{VENT}}{Capacity \cdot V_{ref}} = \frac{\sum_j \left(\dot{m}_j^{VENT,avg} \cdot GWP_j\right)}{Capacity \cdot V_{ref}} \tag{15}$$

$$MGI^{RLIQ} = \frac{\dot{m}_{WtT}^{RLIQ} + \dot{m}_{TtW}^{RLIQ}}{Capacity \cdot V_{ref}} = \frac{PW^{RLIQ,avg} \cdot SFC^{RLIQ} \cdot \left\{ LCV_k \cdot GHG_k^{WtT} + \sum_j \left(CF_j \cdot GWP_j\right) \right\}}{Capacity \cdot V_{ref}} \tag{16}$$

3.2. Case descriptions

The overall cargo handling process was simulated by interconnecting the CHS with a tank model. In Case 1, the excess BOG in the tank is vented into the atmosphere. As indicated in Fig. 4(a), the venting system begins to operate when the tank pressure reaches the maximum operating pressure.

With the BOG re-liquefaction system (Cases 2 and 3), some of the BOG is re-liquefied and returned to the tank. This study assumes two different operating strategies. In Case 2 of continuous operation as

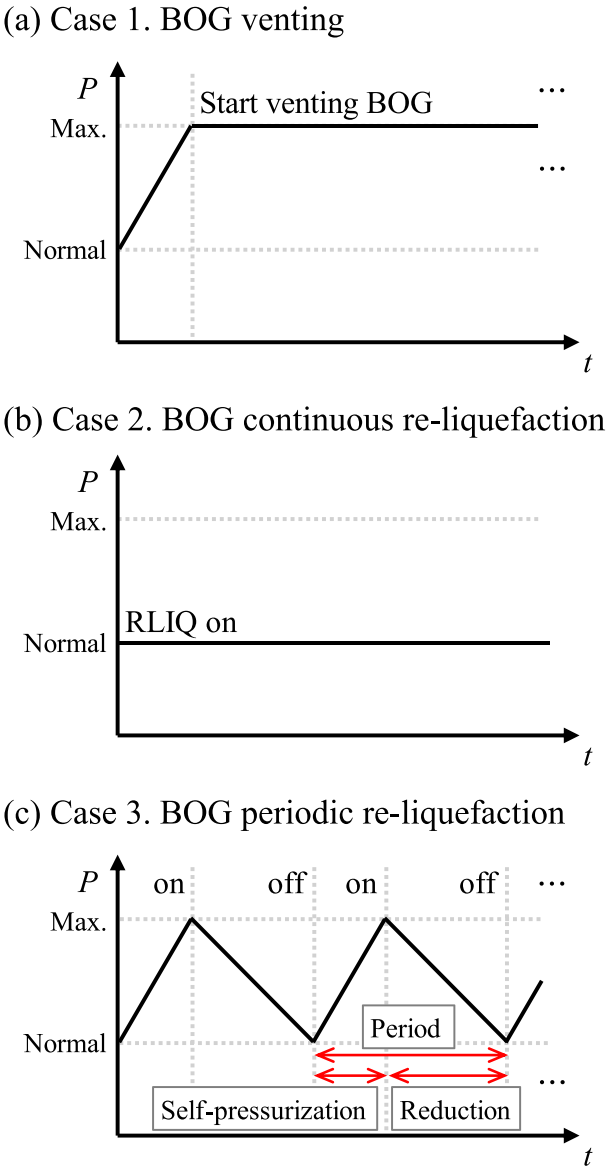


Fig. 4. Example of pressure variation of each case.

shown in Fig. 4(b), the BOG re-liquefaction system is operated continuously to keep the normal operating pressure in the tank during the voyage. In Case 3 of periodic operation as shown in Fig. 4(c), the BOG re-liquefaction system operates in periodic cycles, turning it on and off with pressure variations. One period cycle is decomposed to a period of self-pressurization and pressure reduction. The period of self-pressurization means the period where the pressure increases from normal operating pressure due to the heat ingress without the operation of BOG re-liquefaction system. When the pressure in the tank reaches the maximum operating pressure, the re-liquefaction system is turned on and the pressure in the tank starts to reduce until the tank pressure reaches the normal operating pressure, which is defined as the pressure reduction period. The BOG inflow to the re-liquefaction system exists only during the period of pressure reduction.

3.3. Specifications and initial conditions of cargo tank

Specifications used in this study are listed in Table 3. The target cargo is LCO_2 at the normal operating pressure of 7.5 bar containing 0.2 mol% N_2 [24]. An LCO_2 carrier of 100,000 m^3 (20 × 5000 m^3) is assumed as the target carrier, which is operated at an initial filling level of 95 % on voyage days [16,25]. The target tanks are type C horizontal tanks with elliptical heads in a 2:1 ratio and operate in the range of 6.5–8.5 bar [26]. The insulation performance of the target tank is assumed to satisfy 0.15 % of the initial boil-off rate (BOR_0) per day [27].

As shown in Table 4, the initial cargo conditions that satisfy the specification listed in Table 3 can be determined. The vapor fraction is set to satisfy an initial filling ratio of 95 %, assuming that the vapor and liquid in the tank are in equilibrium. The initial temperature of CO_2 in the tank is estimated based on the normal operating pressure and vapor fraction. The total number of moles of cargo is calculated to ensure that the mixture volume of the liquid and vapor is 100,000 m^3. Calculations are performed using Aspen HYSYS v14, where the Peng–Robinson equation of state (EOS) is used.

3.4. Modelling of cargo tank

As shown in Fig. 5, the target tank can be simplified to estimate the heat and mass transfer. The heat entering the tank from outside through the outer wall (Q_{tot}) is calculated using Eq. (17) and assumed to be constant during voyage.

$$Q_{tot} = BOR_0 \cdot \rho \cdot L_v \cdot V_{tot} \tag{17}$$

where ρ is density of cargo, L_v is latent heat of vaporization and V_{tot} is volume of total tank.

The heat calculated using Eq. (17) is divided into the vapor and liquid parts of the tank. Therefore, the sum of the heat entering the vapor and liquid is constant, as shown in Eq. (3). The ratio of the division of heat between the vapor and liquid is defined using the heat distribution factor (f) in Eq. (18). The heat distribution factor in this study is estimated from the ratio of the temperature difference between the inner

Table 3
Specifications of the target LCO_2 cargo tank.

Item	Unit	Value	Reference
Cargo composition | CO_2	mol%	99.8	[24]
Cargo composition | N_2	mol%	0.2	[24]
Cargo capacity	m^3	100,000 (5000 m^3 × 20 EA)	[16] [25]
Tank type	–	C type horizontal tank	[16]
Head type	–	2:1 elliptical head	–
Maximum operating pressure	bar	8.5	[26]
Normal operating pressure	bar	7.5	[26]
Minimum operating pressure	bar	6.5	[26]
BOR_0	%/day	0.15	[27]

Table 4
Initial conditions of target cargo.

Item	Unit	Value
Pressure	bar	7.5
Vapor fraction	–	0.0009
Temperature	°C	−50.9
Mole	kgmole	2,491,252

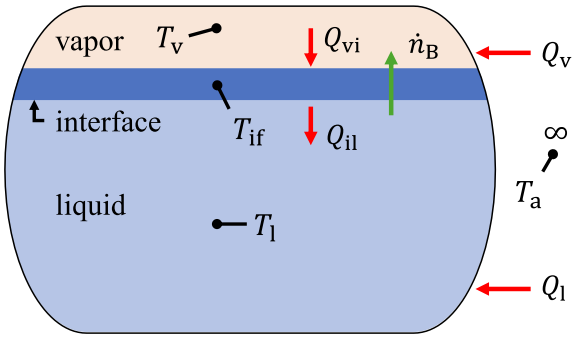


Fig. 5. Computational domain of modified TMZM for LCO_2 cargo tank.

liquid (T_l), vapor (T_v), and ambient temperature (T_a) because the driving force of the heat transfer is the temperature difference.

$$f = \frac{q_v}{q_l} = \frac{T_a - T_v}{T_a - T_l} \tag{18}$$

The heat transfer among vapor, interface, and liquid is calculated as Eqs. (19) and (20). The heat transfer coefficient can be calculated as Eq. (21) using the Nusselt number [28]. The interfacial temperature (T_{if}) is assumed to be the bubble point temperature of the liquid in the tank.

$$Q_{vi} = \frac{Nu_{vi} \cdot k_{th,v}}{L_c} A_i \left(T_v - T_{if}\right) \tag{19}$$

$$Q_{il} = \frac{Nu_{il} \cdot k_{th,l}}{L_c} A_i \left(T_{if} - T_l\right) \tag{20}$$

$$h = \frac{Nu \cdot k_{th}}{L_c} \tag{21}$$

where, k_{th} is thermal conductivity, A is heat transfer area, and L_c is characteristic length.

This study uses Eq. (22) from Holman et al. [29] and Nellis et al. [28] to calculate the Nusselt number for vapor-interface heat transfer.

$$\overline{Nu}_{vi} = \begin{cases} 0.27 \times Ra^{0.25} & \text{at } T_{if} \le T_v \\ \left(\overline{Nu}_{vi,lam}^{10} + \overline{Nu}_{vi,turb}^{10}\right)^{1/10} & \text{at } T_{if} > T_v \end{cases} \tag{22}$$

The Holman et al. [29] equation was used when the vapor temperature was higher than that of the interface (typical case), whereas the Nellis et al. [28] equation was used when the vapor temperature was lower than the interfacial temperature. This approach accounts for both laminar and turbulent effects because the colder fluid on the top increases the convection. The Nusselt numbers for laminar and turbulent flows were respectively calculated using Eqs. (23) and (24).

$$\overline{Nu}_{vi,lam} = 1.4 \left[\ln\left(1 + \frac{1.4}{0.835 C_{lam} Ra_{L_c}^{0.25}}\right) \right]^{-1} \tag{23}$$

$$\overline{Nu}_{vi,turb} = C_{turb,U} Ra_{L_c}^{1/3} \tag{24}$$

The interface-liquid heat transfer was determined using Eq. (25) from Nellis et al. [28] with the average Nusselt number considered laminar.

$$\overline{\mathrm{Nu}}_{\mathrm{il}} = 2.5\left[\ln\left(1+\frac{2.5}{0.527\,\mathrm{Ra}_{L_c}^{0.2}}\left[1+\left(\frac{1.9}{\mathrm{Pr}}\right)^{0.9}\right]^{2/9}\right)\right]^{-1} \quad (25)$$

As shown in Fig. 6, a simulation model is built to estimate thermodynamic tank behavior. The properties of the vapor, interface, liquid, filling ratio, and f factor of the tank are obtained based on the initial conditions of initial pressure (P_0), filling ratio (FR_0), heat entering through the wall (Q_{tot}), and tank geometry (V_{tank} and $(L/D)_{\mathrm{tank}}$). The interface and surface areas of the vapor and liquid were determined based on the tank geometry and filling ratio. The heat entering the vapor and liquid and the heat exchanged at the interface can be calculated using these values. Subsequently, the property change after heat is injected is estimated, and the pressure that makes the volume of the vapor and liquid equal to the tank volume is determined. This process is repeated until the end of the voyage. The thermal properties of CO_2 are estimated using the Peng–Robinson EOS in Aspen HYSYS v14, and the tank model is calculated using MATLAB.

3.5. Modelling and optimization of BOG re-liquefaction system

As shown in Fig. 7, the re-liquefaction system is designed based on the conventional BOG re-liquefaction process with a propane refrigerant using a vapor compression cycle. The propane refrigerant is compressed and cooled (R4), and expanded via J–T valve to make low temperature vapor-liquid propane stream (R5). The excess BOG is compressed and cooled using cooling water and the stream 4 is further cooled by the propane refrigerant (R5) through the heat exchanger. Then the cooled stream 5 is expanded through the J–T valve, and the partially liquefied BOG stream is injected into the LCO_2 tank.

Table 5 lists specifications used in re-liquefaction system simulations. The compressors have an adiabatic efficiency of 85 % and maximum pressure ratio of 3.5. The pressure drop in heat exchangers is ignored and the minimum temperature difference is designed to be 2 °C [30]. The process streams can be cooled down to 30 °C by using cooling water. The lowest temperature of CO_2 should be higher than −55 °C to prevent dry ice formation.

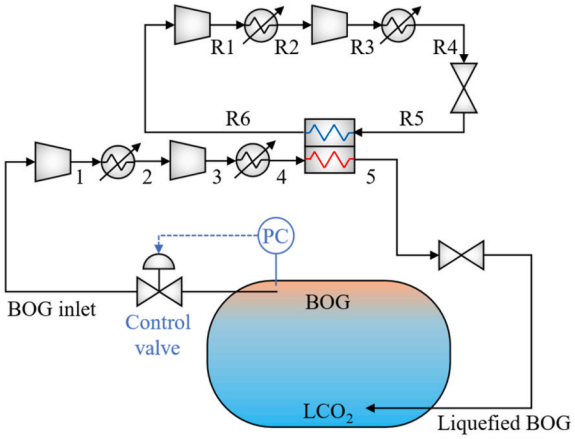


Fig. 7. Process flowchart of BOG re-liquefaction cycle.

Table 5
Specification of re-liquefaction cycle.

Item	Unit	Value
Adiabatic efficiency of compressor	%	85
Pressure drops in the heat exchanger	bar	0
Minimum temperature difference	°C	2
Cooled temperature by cooling water	°C	30
Maximum compressor pressure ratio	–	3.5
Lowest temperature of stream	°C	−55

The re-liquefaction system is optimized to minimize the average power consumed by the BOG re-liquefaction system ($\mathrm{PW}^{\mathrm{RLIQ,avg}}$) where the voyage duration is replaced by the period. The PSO algorithm is used to minimize the objective function and the process variables and boundaries used in the optimization are listed in Table 6. The constraints are provided based on specifications listed in Table 5.

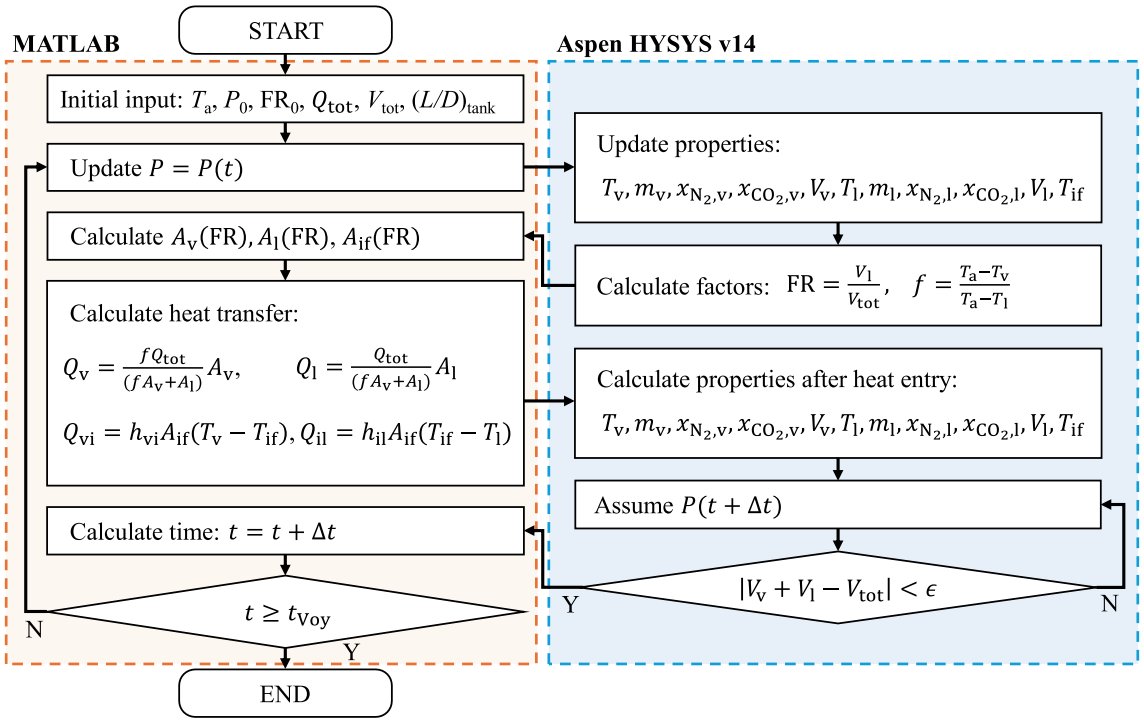


Fig. 6. Flowchart for estimation of tank behavior.

Table 6
Process variables and boundaries for optimizing the re-liquefaction system.

Item	Unit	LB	UB
P of stream 1	bar	8.5	29.75
P of stream 3	bar	8.5	70
T of stream 5	°C	−40	−25
P of stream R1	bar	3.09	10.08
P of stream R5	bar	1.1	3.3
$\dot{n}$ of stream BOG inlet	kgmole/h	300	600

4. Results and discussion

4.1. Self-pressurization of the LCO_2 cargo tank

Fig. 8 shows the estimated pressure and temperature in the LCO_2 tank during 25 days of voyage without CHS. The tank pressure starts to increase with an increase in temperature and BOG generation because of the external heat input. Initially, the pressure rises steeply because there is no temperature difference initially between the vapor and liquid and no interfacial heat transfer occurs. Further, condensation occurs at the interface with an increase in the temperature difference between layers, reducing the rate of pressure increase. In the simulation, a maximum operating pressure of 8.5 bar is reached in 3 days, and the estimated pressure increases to 10.6 bar over 25 days without CHS. Therefore, a strategy to maintain the cargo tank within the operating pressure is essential for voyages longer than three days.

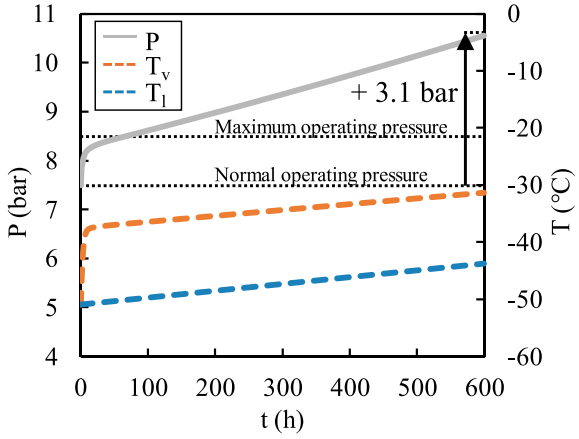


Fig. 8. Tank pressure and temperature over time without the CHS.

4.2. BOG venting (Case 1)

In Case 1, the BOG is vented from the LCO_2 carrier at the maximum operating pressure to prevent overpressure in the tank. Fig. 9 shows the amount of vented CO_2 BOG over time to maintain the tank pressure within 8.5 bar. The tank vents the BOG when the pressure in the tank reaches 8.5 bar. The mole fraction of CO_2 in the BOG (y_{CO_2}) increases as venting begins until the liquid in the tank is saturated, and therefore, the amount of CO_2 contained in the vented BOG ($\dot{n}_{CO_2}^{VENT}$) also increases. Simultaneously, the temperature of the liquid in the tank (T_l) increased gradually, reaching the bubble point temperature (T_{bubble}) at 13 days after departure. When this occurs, the liquid in the tank starts to vaporize and the amount of BOG that needs to be vented ($\dot{n}_{CO_2}^{VENT}$) increases steeply.

The total amount of CO_2 venting over 25 days ($n_{CO_2}^{VENT,tot}$) is estimated as 25,642 kgmole, which is 1.03 % of the total amount of initial CO_2 in the tank. Of the total vented CO_2, the amount vented in the first 13 days is 15.6 %, and the remaining 84.4 % is vented in the next 12 days.

4.3. BOG continuous re-liquefaction (Case 2)

Table 7 lists the optimized re-liquefaction operating conditions based on the steady-state optimization. The BOG re-liquefaction system in Case 2 is operated under optimized operating conditions, excluding the flow rate of the stream BOG inlet. In Case 2, the re-liquefaction flow rate is determined for each unit of time to maintain tank pressure.

The results of PW^{CAE} and $PW^{RLIQ,avg}$ are 786.4 and 540.1 kW, respectively, and the sum of these two values, PW^{AE}, is 1326.5 kW. This is analogous to the power output of the auxiliary engines installed on the gas carriers in operation listed in Table 8. Since the ship's size in this study are larger than those used for comparison, it is expected that it will be equipped with larger auxiliary engine, which will provide sufficient power to run the re-liquefaction system.

Fig. 10 shows simulation results for the continuous operation of the

Table 7
Optimization results of the re-liquefaction system.

Item	Unit	Value
P of stream 1	bar	26.72
P of stream 3	bar	31.47
T of stream 5	°C	−25.62
P of stream R1	bar	5.20
P of stream R5	bar	1.84
$\dot{n}$ of stream BOG inlet	kgmole/h	352.1
$PW^{RLIQ,avg}$	kW	541.7

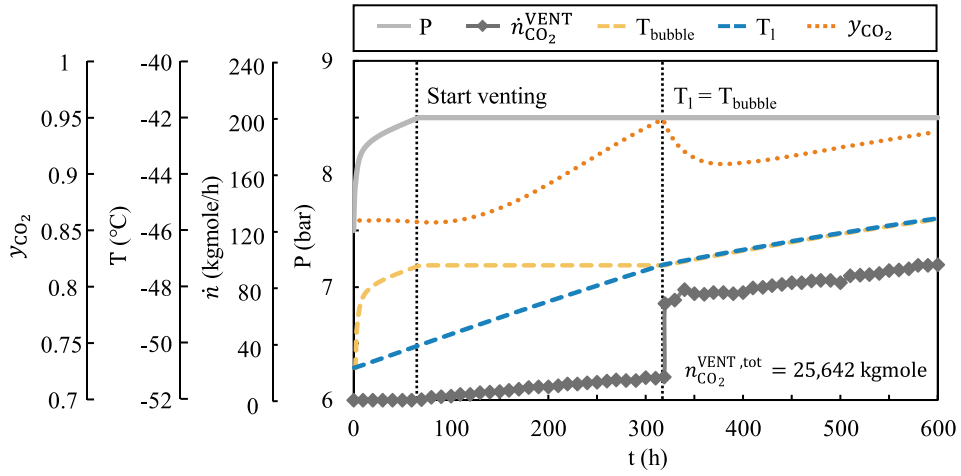


Fig. 9. Vented CO_2 and tank temperature over time for Case 1.

Table 8
Engine specifications of the gas carrier [31].

Ship name	AXIS RIVER	GAS JUSTESEN	This study
Size	86,919 m^3	85,998 m^3	100,000 m^3
Main engine	MAN B&W 6G60ME-C10.5-LGIP	MAN B&W 6G60ME-C10.5-HPSCR	Estimated based on Eq. (11)
PW^{ME}	12,150 kW	11,900 kW	16,092 kW
Auxiliary engine	Taiyo Electric Co., Ltd/FE 653B-8	HD Hyundai Electric, HFC7 564-08P	Estimated based on Eqs. (12) and (13)
PW^{AE}	1300 kW	1550 kVA	1327 kW

BOG re-liquefaction system with the re-liquefaction flow rate controlled to maintain a tank pressure of 7.5 bar. The required flow rate increases sharply within the first 10 h to a maximum required flow rate of 347.1 kgmole/h. Then, it converges to 301.9 kgmole/h, whereas the tank pressure and temperature remained almost constant. In Case 2, the average power consumption ($PW^{RLIQ,avg}$) is 569.9 kW, which is 5 % higher than the optimum value in Table 7; however, the average re-liquefaction flow rate ($\dot{n}^{RLIQ,avg}$) is 302.2 kgmole/h, which is 14 % lower than the optimum value in Table 7.

4.4. BOG periodic re-liquefaction (Case 3)

The BOG re-liquefaction system in Case 3 is operated under optimized conditions listed in Table 7. Fig. 11(a) shows the tank pressure and temperature variation when operating the re-liquefaction system by switching it on and off during the voyage of 25 days. The period of self-pressurization from 7.5 to 8.5 bar and the reduction to 7.5 bar by the re-liquefaction system is repeated twice over 25 days, 12 days apart. The temperatures of the BOG and LCO_2 in the tank increases when the re-liquefaction is not in operation and then decreases with the re-liquefaction. In Case 3, the average power consumption ($PW^{RLIQ,avg}$) is 540.1 kW, which is similar to the optimum value. Despite the same operating conditions, it is beneficial to operate periodically to reduce average power consumption by 5 % compared to Case 2.

The PT diagram in Fig. 11(b) shows the change in the thermodynamic states of the LCO_2 in the tank over one period of re-liquefaction. State (1) is the initial starting condition, and then, the tank pressure and temperature increase by heat influx, reaching state (2) at a maximum operating pressure of 8.5 bar. As shown in Fig. 11(b), the pressure in the tank drops faster and meets the bubble point line in state (3) when the re-liquefaction operation starts. The process of depressurization from (2) to (3) is fast, and it occurs within 2 h. The process of depressurization from (3) to (1) is relatively slow because the liquid becomes saturated at (3) and evaporation occurs from this point.

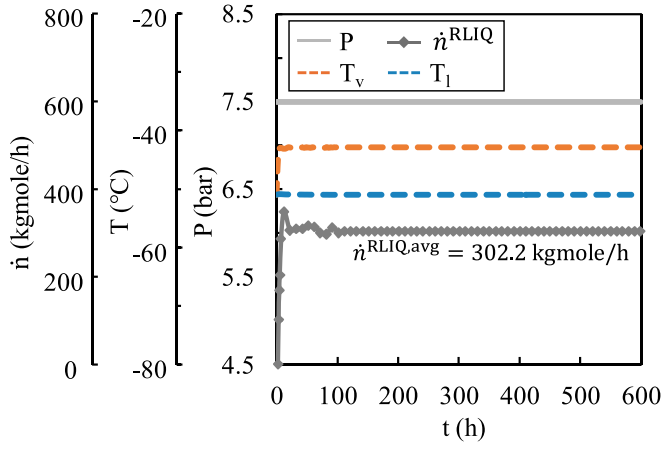


Fig. 10. Molar mass flow of BOG inlet and tank temperature over time for Case 2.

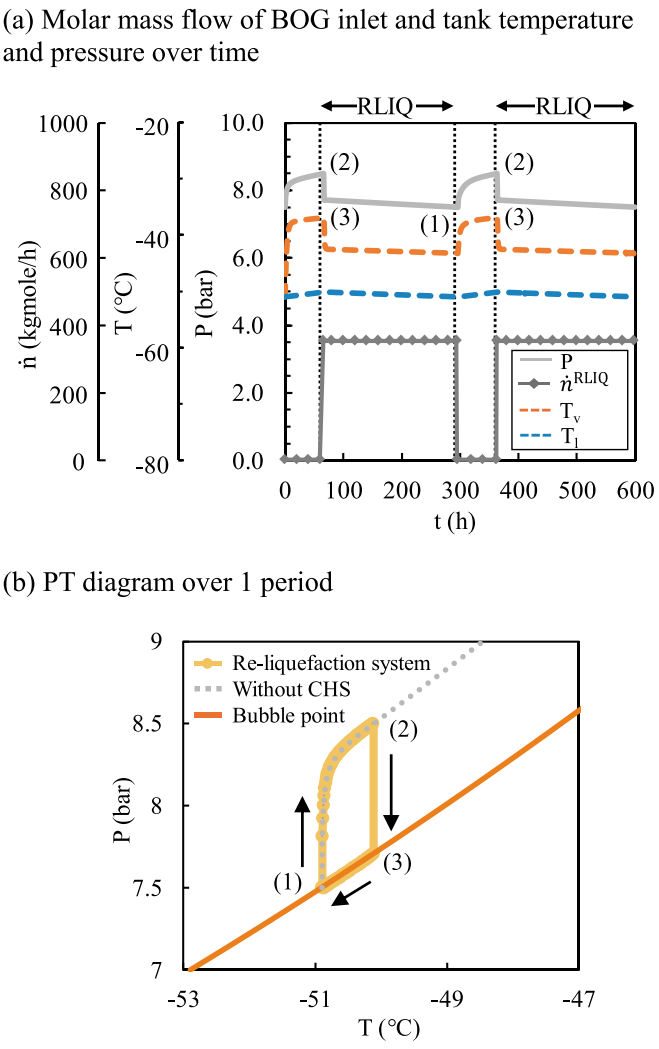


Fig. 11. Tank behavior variation over time for Case 3.

4.5. MGI analysis

Fig. 12 shows the MGI of Case 1 (BOG venting), Case 2 (BOG continuous re-liquefaction), and Case 3 (BOG periodic re-liquefaction) for 25 voyage days for 10,797 nautical miles with a reference velocity of 18 knots. To compare the calculated MGI with the EEDI regulation set out by the IMO, Fig. 12 shows the reference line for the target LCO_2 carrier and the required EEDI based on the reduction factor (X) as solid lines. The result indicates that the original EEDI ($MGI^{ME}_{TtW} + MGI^{CAE}_{TtW}$) without consideration of additional GHG emissions from venting or reliquefaction in both cases are around 3.30 $g_{CO2eq}/(t \cdot nm)$, satisfying the 40 % reduction target of EEDI. However, when the CO_2 vent and WtW GHG emissions from the LCO_2 carrier are considered, the MGI of Case 1increases to 5.36 $g_{CO2eq}/(t \cdot nm)$, only satisfying 5 % reduction. Especially in Case 1, the increase in MGI caused by the vented CO_2 (MGI^{VENT}_{TtW}) is significant, resulting in an increase of 0.964 in the MGI. That is, as expected, CO_2 vent is not a good solution for long distance voyage. For Cases 2 and 3 with the re-liquefaction system, the MGI increased by the re-liquefaction system (MGI^{RLIQ}_{TtW}) is 0.109 and 0.104, respectively. Accordingly, Cases 2 and 3 exhibits around 16 % lower to 4.45 $g_{CO2eq}/(t \cdot nm)$ than that of Case 1, which satisfy the 20 % reduction target.

The BOG re-liquefaction system is not dominant in all voyage scenarios. Fig. 13 shows the MGI caused by the CHS in each case as a function of voyage days. In Case 1, the amount of vented CO_2 increases

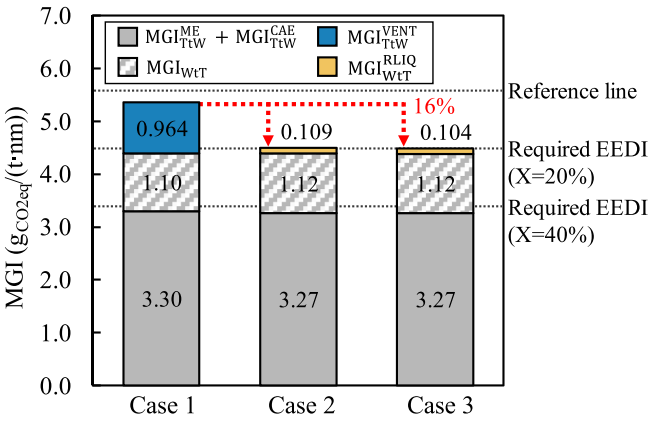


Fig. 12. MGI of each case for 25 days of voyage.

gradually over time and then steeply around day 13 as shown in Fig. 9; therefore, the MGI^{CHS} also increases gradually, and a steep increase can be seen at 13 voyage days. In Case 2, MGI^{CHS} is close to a constant of 0.146 g_{CO2eq}/(t·nm), regardless of the voyage days due to the continuous operation. In Case 3, MGI^{CHS} is initially zero because there is no need to operate re-liquefaction. As the pressure reaches the maximum pressure of 8.5 bar and re-liquefaction is operated, the MGI^{CHS} increases until 13 voyage days. Then, it decreases at the start of the second period and increases again at the start of the pressure reduction period to reach a constant MGI^{CHS} of 0.138 g_{CO2eq}/(t·nm).

The intersections of the MGI for each case show the dominant cargo handling strategy. At a short distance of three days or less, the tank can withstand pressure without any CHS. Even up to 13 days of voyages, venting CO_2 (Case 1) has lower MGI than those of re-liquefaction cases (Cases 2 and 3) because the LCO_2 in the tank was subcooled, and only a small amount of venting is required to maintain tank pressure within the operating pressure. However, for longer voyages of 13 days or more, Cases 2 and 3 with the re-liquefaction system are preferable because the vented CO_2 in Case 1 increases considerably faster.

The MGIs of Cases 2 and 3 after 13 days are at a similar level. However, Case 3 has a lower MGI than Case 2 for a shorter voyage of 10 days or less. Therefore, for LCO_2 carriers with a wide range of voyage days, the periodic re-liquefaction can be beneficial for enabling more flexible operation to cover shorter voyages, although the strategy may require a more complex operation.

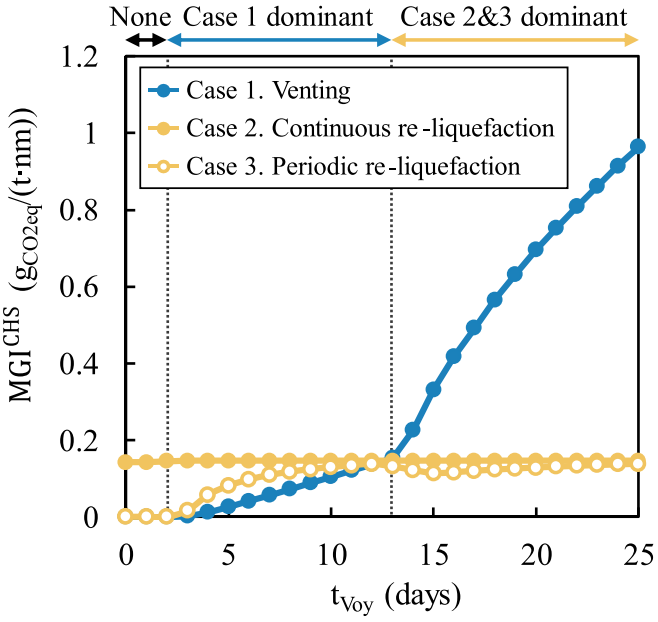


Fig. 13. MGI of CHS and proposed cargo handling strategy by voyage days.

4.6. Sensitivity analysis with change of BOR_0

Because the design specification of BOR_0 can affect the results fundamentally, sensitivity analysis was performed with change of BOR_0 from 0.10 to 0.20 %/day. Table 9 lists the results of simulating 25 days of self-pressurization and BOR venting (Case 1), BOR continuous re-liquefaction (Case 2), and BOR periodic re-liquefaction (Case 3) based on different BOR_0. When BOR_0 increases from 0.10 to 0.20 %/day, the time to reach the maximum operating pressure of 8.5 bar is extremely shortened to 18 h after departure from 191 h after departure in 0.10 %/day. For the BOG venting system (Case 1), the vented amounts of CO_2 are 0.357, 1.126, and 2.046 ton of CO_2 for the BOR_0 of 0.1, 0.15, and 0.2 %/day, respectively. For continuous operation of the re-liquefaction (Case 2), the required molar flow rates of re-liquefied BOG are an average of 198.5, 299.4, and 396.2 kgmole/h, respectively. For periodic operation of the re-liquefaction (Case 3), it is operated at fixed flow rates of 250, 352, and 410 kgmole/h, respectively. The larger the BOR_0, the longer the system operated, with total operating times of 409, 467, and 582 h, respectively.

Fig. 14 shows the MGI for each BOR_0 and CHS case when the voyage days are 25 days. When BOR_0 is 0.1 %/day, the MGI^{CHS} is 0.303, 0.096, and 0.083 g_{CO2eq}/(t•nm) for Cases 1, 2, and 3, respectively. When BOR_0 is 0.2 %/day, MGI^{CHS} is 1.763, 0.192, and 0.192 g_{CO2eq}/(t•nm) for Cases 1, 2, and 3, respectively. The MGIs of Case 2 and 3 reduced by only 5 % comparing to Case 1 for BOG_0 of 0.1 %/day, but the MGIs of re-liquefaction systems (Case 2 and 3) are reduced by 27 % comparing to venting case (Case 1) when BOG_0 is 0.2 %/day. That is, with larger BOG_0, the advantage of re-liquefaction system increases, because of the increased amount of BOG.

Fig. 15 shows MGI^{CHS} according to BOG_0, CHS, and voyage days. For all three BOR_0s, in Case 1 (venting), MGI tends to increase slowly with voyage days and rise steeply after a certain point. In Case 2 (continuous re-liquefaction), it remains at a constant level except when voyage days are very short. In Case 3 (periodic re-liquefaction), fluctuations corresponding to the re-liquefaction period can be observed. Case 3 shows a relatively lower MGI than Case 2 over the entire voyage days when BOR_0 is 0.1 and 0.15 %.day. However, at BOR_0 of 0.2 %/day, the MGI of Cases 2 and 3 reaches is at a similar level. Case 3 is a strategy that reduces MGI by lowering total power consumption through shortening the re-liquefaction operation time instead of increasing the re-liquefaction size, compared to Case 2. However, at BOR_0 of 0.2 %/day, where pressure rises rapidly, the time when the system is turn off for a short time, so it is difficult to gain an advantage.

Table 9
Simulation results of each case according to BOR_0 (%/day) of 0.10, 0.15, 0.20.

Case	Item	Unit	BOR_0 (%/day)		
			0.10	0.15	0.20
Self-pressurization	Time to reach maximum operating pressure of 8.5 bar	h	191	66	18
	Pressure after 25 days	bar	9.5	10.6	11.7
BOG venting (Case 1)	Total vented CO_2 during 25 days	ton	0.357	1.126	2.046
Continuous re-liquefaction (Case 2)	Average re-liquefied molar flow	kgmole/h	198.5	299.4	396.2
	Average power consumption	kW	374.3	565.1	749.9
Periodic re-liquefaction (Case 3)	Re-liquefied molar flow	kgmole/h	250.0	352.1	410.0
	Operating hours	h	409	467	582
	Average power consumption	kW	325.5	541.0	749.8

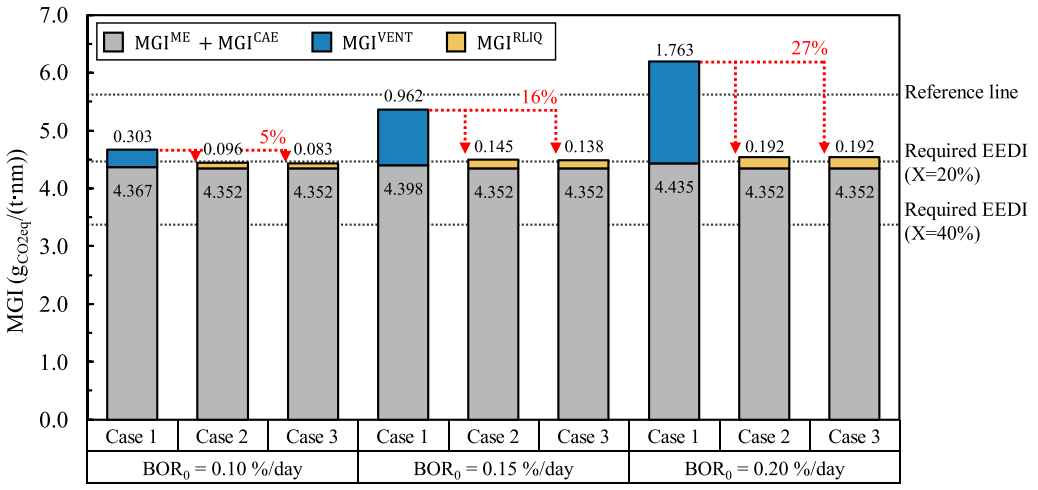


Fig. 14. Impact of BOR_0 on MGI of each case for 25 days of voyage.

BOR_0	0.10 %/day			0.15 %/day			0.20 %/day		
CHS	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3
0	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	0.000	0.105	0.000	0.000	0.147	0.000	0.011	0.050	0.050
2	0.000	0.102	0.000	0.000	0.147	0.000	0.040	0.124	0.124
3	0.000	0.100	0.000	0.002	0.146	0.017	0.075	0.149	0.149
4	0.000	0.099	0.000	0.012	0.146	0.057	0.101	0.161	0.161
5	0.000	0.098	0.000	0.026	0.145	0.081	0.127	0.168	0.168
6	0.000	0.098	0.000	0.041	0.145	0.098	0.152	0.173	0.173
7	0.000	0.097	0.000	0.057	0.145	0.109	0.177	0.177	0.177
8	0.000	0.097	0.001	0.073	0.145	0.118	0.201	0.179	0.179
9	0.003	0.097	0.014	0.089	0.145	0.124	0.225	0.182	0.182
10	0.007	0.097	0.025	0.105	0.145	0.130	0.247	0.183	0.183
11	0.012	0.097	0.034	0.121	0.145	0.134	0.419	0.185	0.185
12	0.019	0.097	0.041	0.137	0.145	0.138	0.573	0.186	0.186
13	0.026	0.096	0.047	0.153	0.145	0.131	0.705	0.187	0.187
14	0.033	0.096	0.053	0.235	0.145	0.122	0.826	0.187	0.187
15	0.041	0.096	0.058	0.332	0.145	0.114	0.937	0.188	0.188
16	0.049	0.096	0.062	0.418	0.145	0.116	1.039	0.189	0.189
17	0.058	0.096	0.065	0.493	0.145	0.120	1.134	0.189	0.189
18	0.066	0.096	0.068	0.565	0.145	0.123	1.225	0.190	0.190
19	0.074	0.096	0.071	0.632	0.145	0.126	1.312	0.190	0.190
20	0.095	0.096	0.074	0.695	0.145	0.129	1.395	0.191	0.191
21	0.144	0.096	0.076	0.754	0.145	0.131	1.474	0.191	0.191
22	0.189	0.096	0.078	0.808	0.145	0.133	1.548	0.191	0.191
23	0.230	0.096	0.080	0.862	0.145	0.135	1.619	0.192	0.192
24	0.268	0.096	0.082	0.914	0.145	0.137	1.691	0.192	0.192
25	0.303	0.096	0.083	0.962	0.145	0.138	1.763	0.192	0.192

t_{voy} (day)

(g_{CO2eq}/(t·nm))

Min.

Max.

Fig. 15. MGI^{CHS} according to BOR_0, CHS, and voyage days.

Table 10 analyzes the point where the MGI^{CHS} of each CHS crossover occurs based on voyage days for each BOR_0. When BOR_0 is 0.1 %/day, no CHS is required up to 8 days because the tank can withstand the pressure. For voyages up to 19 days, the venting system is advantageous, while for voyages exceeding 20 days, the re-liquefaction system is advantageous. Thus, with low BOR_0 and short voyage plan, the re-liquefaction system is not attractive. When BOR_0 is 0.2 %/day, however, the tank reaches maximum pressure in less than a day. Therefore, regardless of voyage days, consideration of cargo handling strategies is essential, and the re-liquefaction system is recommended for voyages of 7 days or longer.

Table 10
Proposed cargo handling strategy by voyage days according to BOR_0.

BOR_0 (%/day)	0.10	0.15	0.20
Nothing required	0–8 days	0–3 days	0
BOG venting system	9–19 days	4–12 days	0–7 days
BOG re-liquefaction system	20–25 days	13–25 days	7–25 days

4.7. *Future work*

This study did not perform a cost-benefit analysis of the BOG handling system because its objective is to highlight the MGI's strengths as the GHG intensity index for ships. However, the high capital expenditure of the re-liquefaction system is a notable barrier to its adoption, so economics is also a key index. The results of the cost-benefit analysis vary depending on factors such as carbon taxes, capture costs, and tank prices, necessitating sensitivity analysis. Future work will present economic criteria for adopting the BOG handling system through cost-benefit analysis and sensitivity analysis.

5. Conclusion

This study suggests a new index of MGI to assess the WtW GHG intensity generated during the entire ship transportation process, and shows the effectiveness of CO_2 re-liquefaction systems quantitatively considering two distinguishable strategies of continuous and periodic

operations. The integrated thermodynamic model combining the cargo tank with the CHS is developed and simulated for 25 days of voyage. The results with BOR_0 of 0.15 %/day show that the pressure reaches 8.5 bar by day 3 and 10.6 bar by day 25 without CHS, so CHS is inevitable for long distance voyage of the LCO_2 carrier. For the voyage of 25 days, BOG venting (Case 1) resulted in 1.03 % cargo loss and MGI of 5.36 g_{CO2eq}/(t·nm), while BOG re-liquefaction (Cases 2 and 3) reduced MGI by 16 % to 4.45 g_{CO2eq}/(t·nm). Based on the day of the voyage, BOG venting (Case 1) has lower MGI for voyages under 13 days because of the low amount of vented CO_2, but after 13 days, Cases 2 and 3 with re-liquefaction system has lower MGI due to the increased amount of vented CO_2. Cases 2 and 3 show similar MGI for long voyages but Case 3 show much lower MGI for voyages of less than ten days. This means that the periodically operated re-liquefaction system can be considerable for LCO_2 carriers with wide range of voyage days because it enables a more flexible operation that can cover shorter voyages. The results of sensitivity analysis show that for lower BOR_0 of 0.1 %/day, no CHS is required up to 8 days and venting system (Case 1) is advantageous up to 19 days, while the re-liquefaction system (Cases 2 and 3) is advantageous for voyages exceeding 20 days. For higher BOR_0 of 0.2 %/day, however, the tank reaches maximum pressure in less than a day. Therefore, consideration of cargo handling system is essential, and the re-liquefaction system (Cases 2 and 3) is recommended for voyages of 7 days or longer.

CRediT authorship contribution statement

Yunju Jeon: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Hongkyoung Shin:** Methodology, Conceptualization. **Jungmin Kim:** Methodology, Investigation. **Ijun Jeong:** Validation, Conceptualization. **Youngsub Lim:** Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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